# Receipt & Invoice Digitizer

AI-Powered Bills Management System

Team- D

Milestone- II

HTML • CSS • JavaScript • Python • SQL • OCR

# Team- D members

Trishala, Sriya, Rithwik, Mastan and Atul

HTML • CSS • JavaScript • Python • SQL • OCR

# Project Overview

## Purpose

A comprehensive web application designed to digitize and manage receipts and invoices efficiently, eliminating the hassle of paper-based record keeping.

## Key Benefits

Automated data extraction using OCR technology

Secure cloud storage for all documents

Easy search and retrieval of financial records

Analytics and reporting capabilities

# Core Features

## User Authentication

Secure registration and login system with password encryption

## Document Upload

Support for images and PDFs with drag-and-drop functionality

## OCR Processing

Automatic extraction of vendor, date, amount, and tax information

## Smart Search

Filter receipts by vendor, date, and invoice number

## Data Management

View, edit, and delete receipts with ease

## Analytics Dashboard

Visualize spending patterns and generate reports

# Technology Stack

## Backend

Flask (Python Web Framework)
SQLite Database
bcrypt (Password Hashing)
Werkzeug (File Handling)
Session-based Authentication

### OCR Integration (Planned)

Tesseract OCR / Google Vision API
Automatic data extraction

## Frontend

HTML5, CSS3, JavaScript
Responsive Design
Modern UI with gradient theme
Drag-and-drop file upload
AJAX for dynamic updates

### Design Features

Custom pink gradient theme
Smooth animations
Clean, modern interface

# Frontend Architecture

## Code Organisation

Clean, modular code structure with separation of concerns:

### HTML Structure

- Semantic HTML5 elements
- Navbar with navigation items
- Centered header with large title
- Card-based layout
- Table for receipts display

### JavaScript Logic

- JWT authentication check
- File upload handling
- Drag & drop implementation
- API integration (fetch)
- Dynamic table rendering

### CSS Styling

- Custom pink gradient background
- Flexbox for layouts
- Hover effects & transitions
- Responsive media queries
- Custom scrollbar styling

### Design

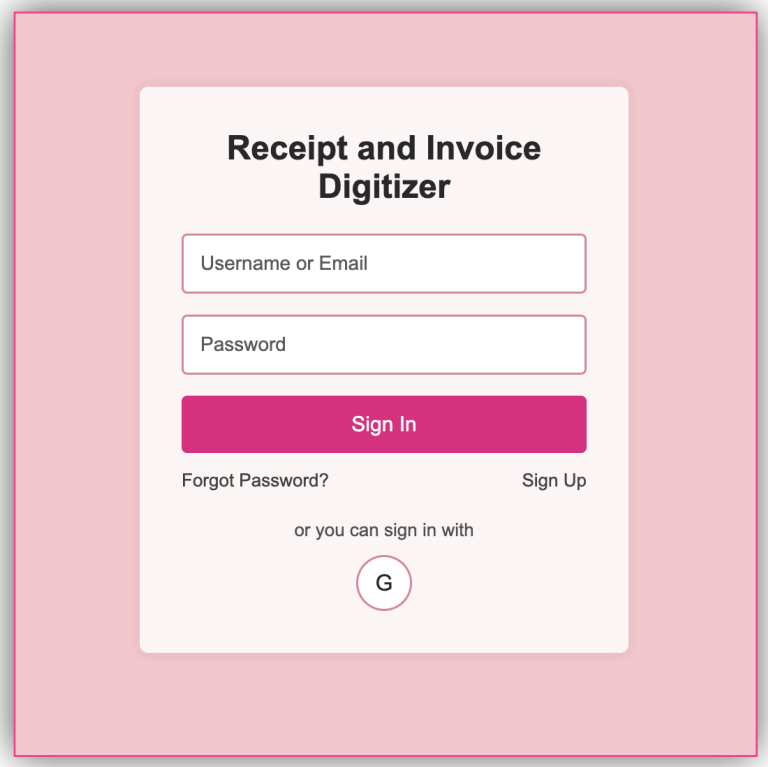
- Custom fonts: DM Sans & Space Grotesk
- Gradient backgrounds: #f5d5d8 → #e8b4b8 → #d4989c
- Rounded corners and glassmorphism effects

# Frontend Structure

## Login and Forgot Password

### Login Design

1. Username or Email field for user identification
2. Password field for secure authentication
3. Sign In button to access the application
4. Forgot Password option for credential recovery
5. Sign Up option for new user registration.

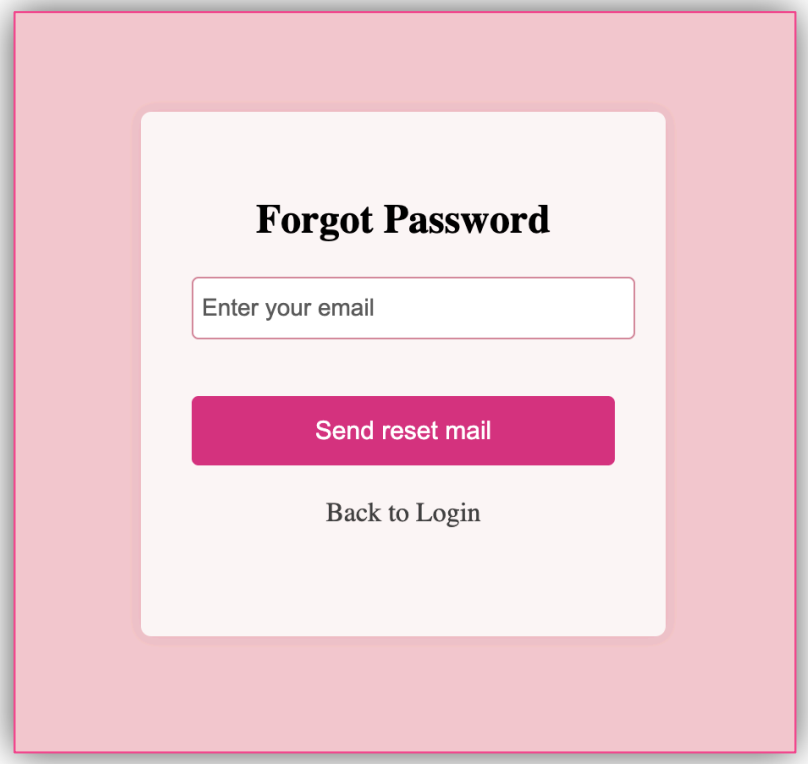


# Frontend Structure

## Login and Forgot Password

### Forgot password Design

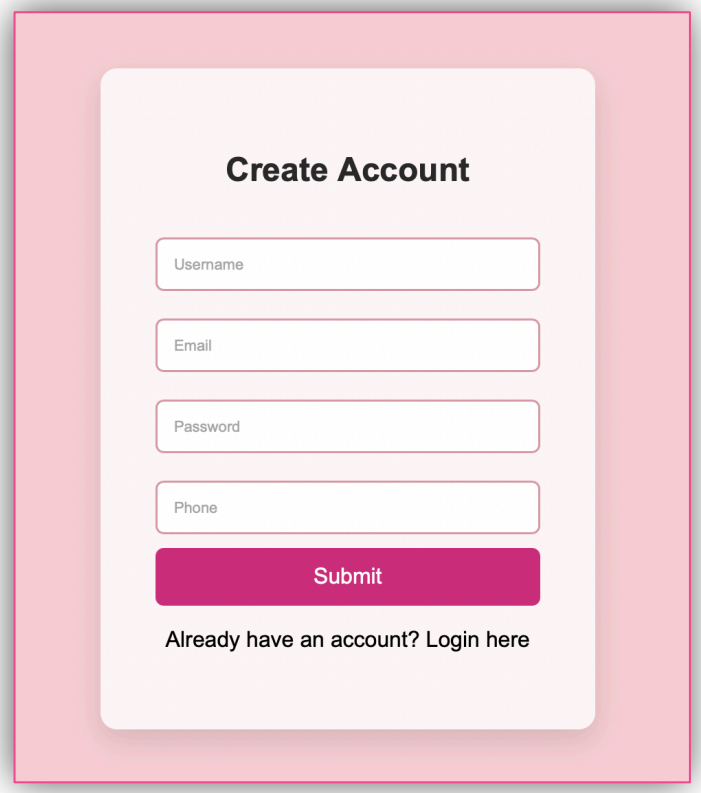
1. The forgot password page has an email input field for verification.
2. It provides an option to send a password reset email.
3. A back-to-login option is included for easy navigation.



# Frontend Structure

## Registration Page

1. The Registration Page provides a clean and user-friendly interface for new users to create an account.
2. The design follows a modern card-based layout with a clear call-to-action button.
3. The form includes proper validation for all input fields, ensuring data integrity and providing real-time feedback to users.
4. The interface offers a seamless transition to login for existing users, reducing friction in the user authentication flow.



# Frontend Structure

## Dashboard

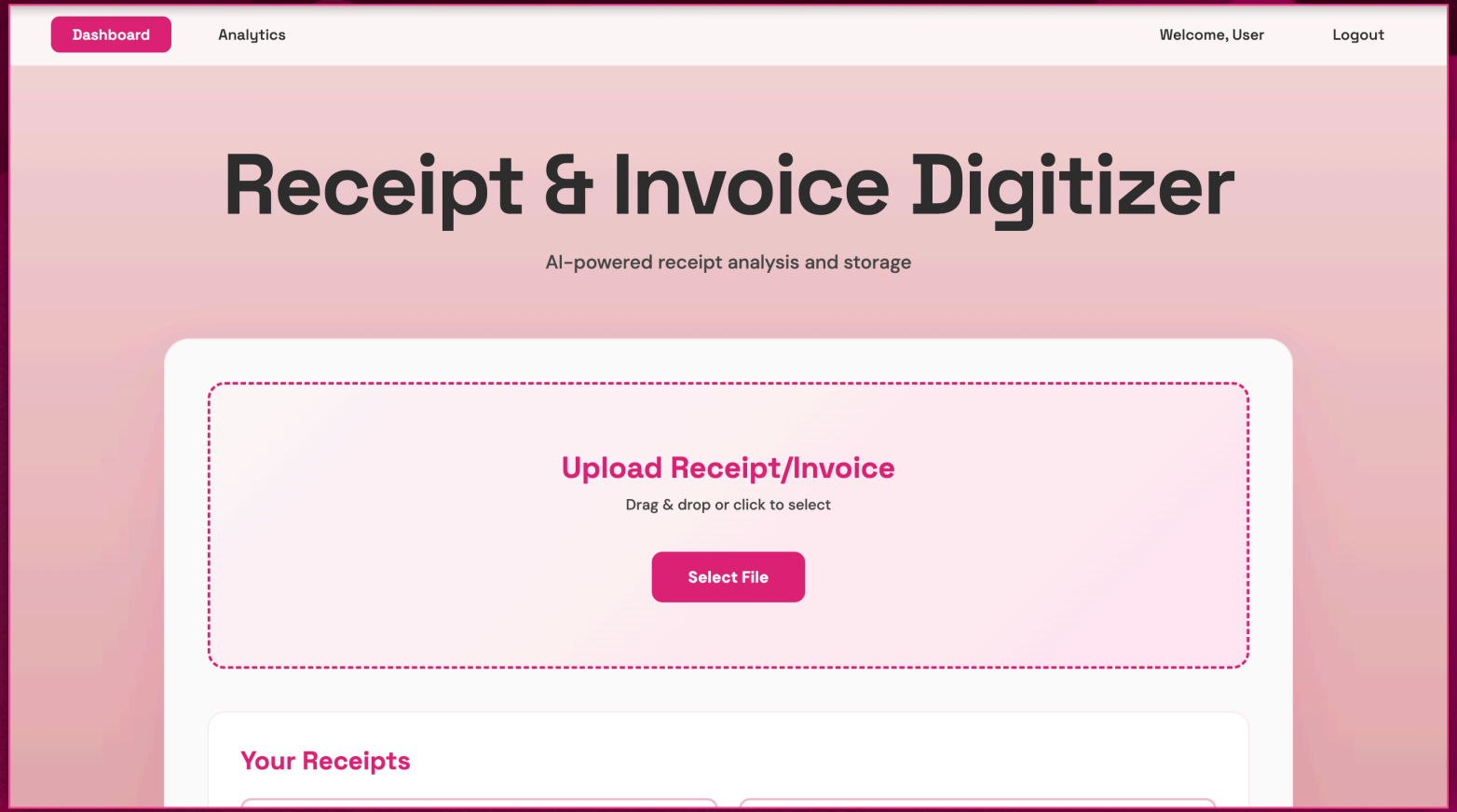
1. **Sticky Navbar**
   Automatically hides on scroll down, reappears on scroll up for better UX

2. **Centered Hero Section**
   Large 80px title with gradient background creates strong visual impact

3. **Upload Zone**
   Dashed border, hover effects, and drag-drop functionality for easy file uploads

4. **Data Table**
   Clean table design with search filters and action buttons for each receipt

# Frontend Structure



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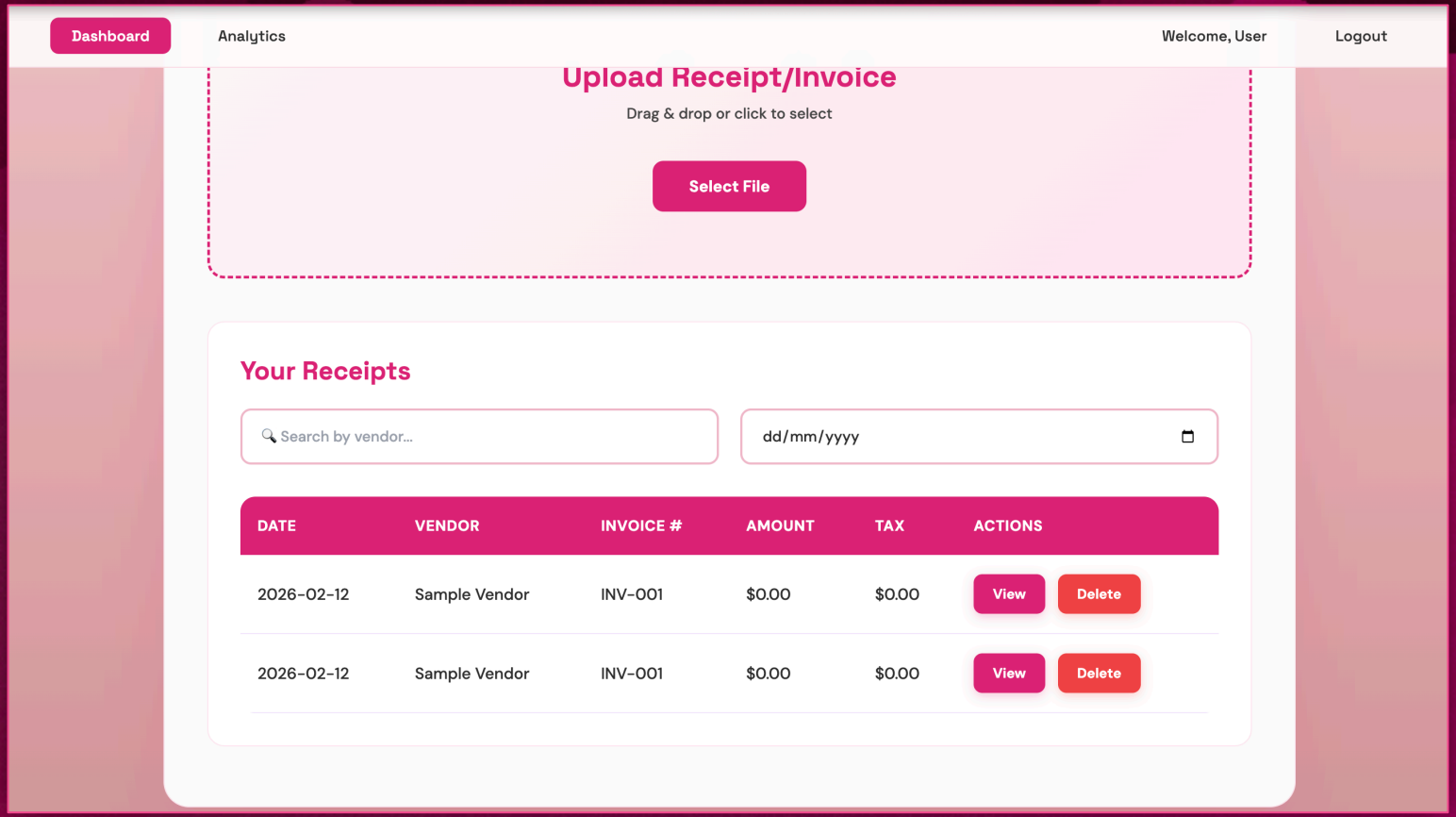

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# Backend Architecture

### Python & Flask

Core development framework for building scalable RESTful services.

### SQLite

Lightweight relational database for structured user information storage.

### bcrypt

Industry-standard password hashing to ensure cryptographic security.

# Backend Development

- Designed core Flask backend architecture
- Developed secure Login/Register endpoints
- Integrated backend logic with HTML forms
- Implemented bcrypt password hashing
- Configured and optimized SQLite storage
- Resolved complex integration bugs

# Authentication Flow

**User Input**
Form submission via Frontend

**API Processing**
Endpoint routing in Flask

**Hashing**
Secure bcrypt encryption

**DB Storage**
SQLite Commit & Validate

**Verification**
Session-based response

# Database Schema

## Users Table

id (Primary Key)
username (Unique)
email (Unique)
password (Hashed)
phone

## Database Features

SQLite database with automatic initialization
Foreign key relationships ensure data integrity
Automatic timestamp tracking for uploads
Efficient indexing on username and email fields

## Receipts Table

receipt_id (Primary Key)
user_id (Foreign Key)
filename
upload_date
vendor_name, invoice_number
date, total_amount, tax

# Security & Data Protection

### 🔒 Password Security
bcrypt hashing with unique salts for each password

### 🛡️ Session Management
Secure session tokens with automatic expiration

### 📁 File Validation
Strict file type and size checks before upload

### 🔐 Access Control
User-specific data isolation and authentication

### 🚫 SQL Injection Prevention
Parameterized queries for all database operations

### ✅ Input Sanitization
Werkzeug secure filename generation

Milestone-2

# OCR Integration

## Automated OCR + Local LLM Structuring Architecture

1

### Drag & Drop Upload

User uploads receipt image or PDF

2

### Flask Upload API

OCRfile via Flask route

3

### Tesseract OCR

Text extraction from receipt image

4

### Gemma 2B (Local LLM)

LLM structures extracted text into JSON

5

### Regex Fallback (Safety Layer)

Regex validates critical fields

6

### SQLite Database + Dashboard

Saved in SQLite
Displayed in Dashboard

# User Workflow

1. Register / Login
2. Upload Receipt
3. OCR Processing
4. Review & Edit
5. Search & Filter
6. View Analytics

# Developments in milestone-2

## OCR development

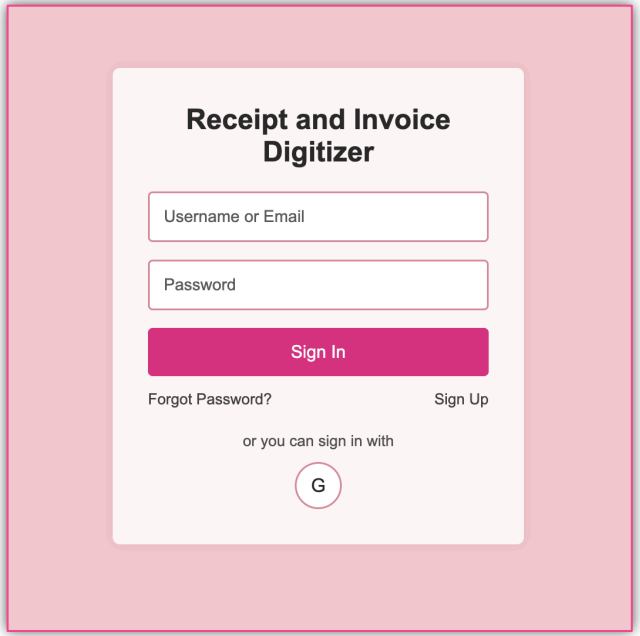
An advancement in the ocr model

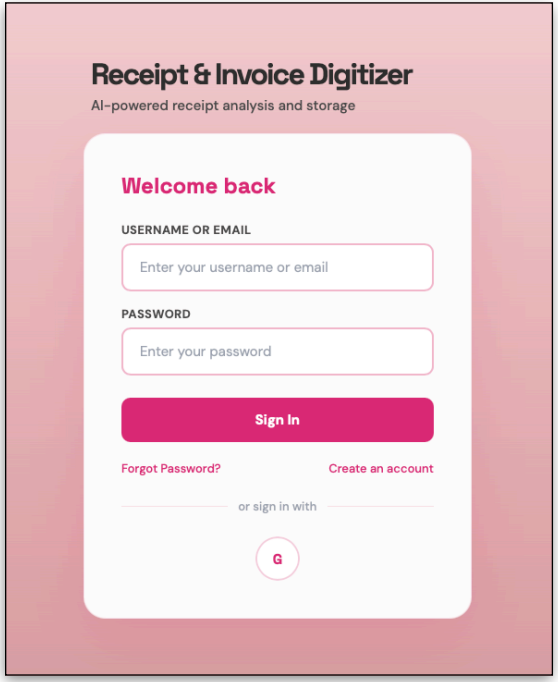
## UI/UX

The older UI had been changed to match the theme

# Frontend Structure

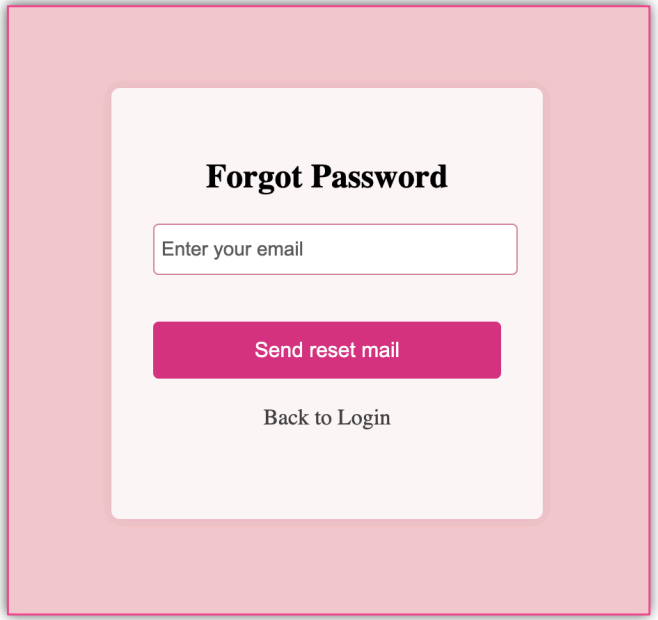
## Login page

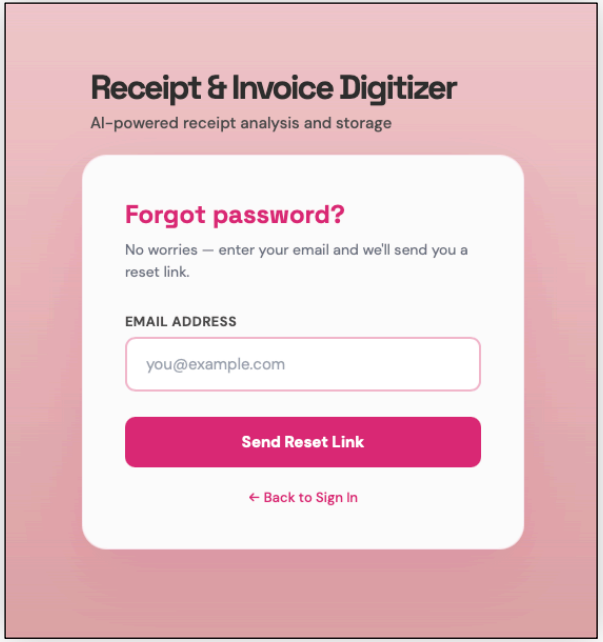


# Frontend Structure

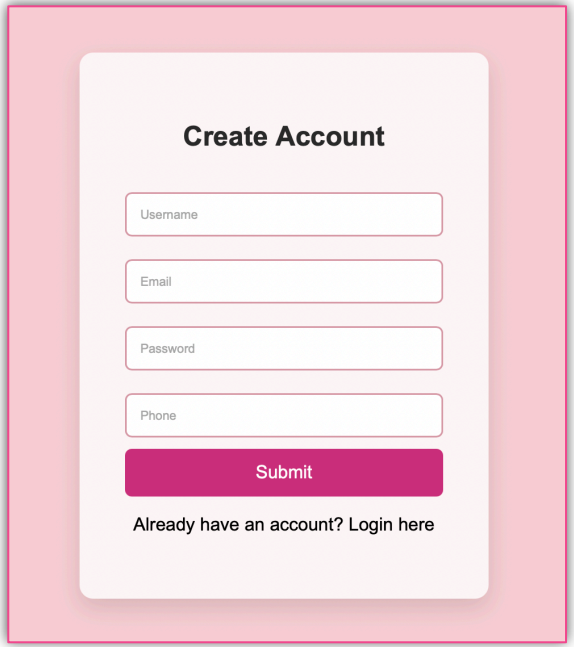
## Forgot Password

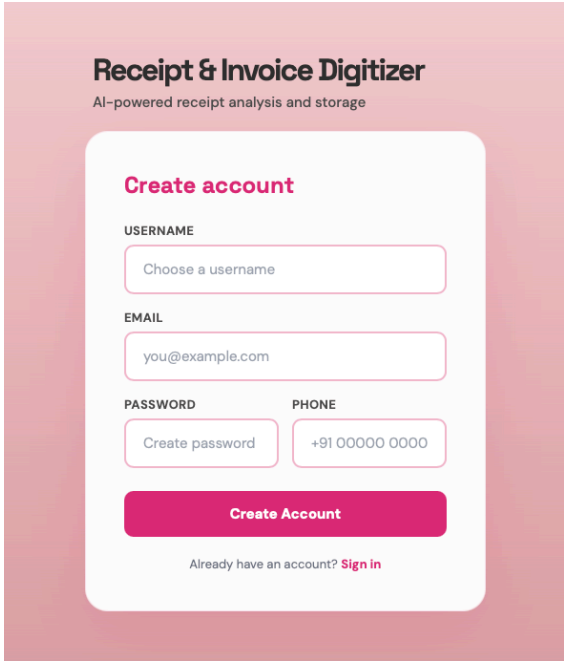


# Frontend Structure

## Registration Page



# Developments in milestone-3

## Google Sign in- Authentication

OAuth 2.0 — users can log in with their existing Google account

## Admin Dashboard

Admin logs in with special credentials to view all users & receipts

## UI/UX, Analytics View

Development in the UI/UX, Line graph tracking daily spending & tax, summary cards

## Multi- Language

Full UI support: English, Telugu, Hindi, Malayalam, Tamil, Kannada

# Google OAuth

## Why?

- No separate password to remember
- One-click login with Google account
- OAuth 2.0 — industry standard security
- Reduces sign-up friction for new users
- Pulls name, email & avatar automatically
- New users get account created instantly

## OAuth 2.0 Flow

G User clicks Sign in with Google

→ Redirect to Google consent screen

✓ User approves email & profile access

↩ Auth code returned to Flask callback

🏠 Session created, user goes to Dashboa

# Admin Dashboard

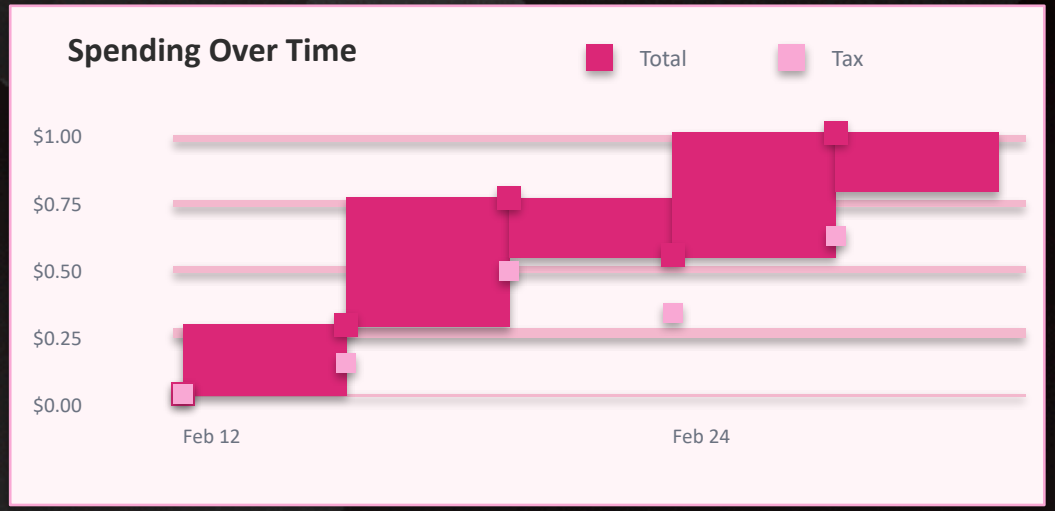
## Admin Can

- Log in with dedicated admin credentials View all registered users & profiles
- See every receipt across all accounts
- View platform-wide analytics & totals
- Use the same familiar dashboard UI
- Regular users cannot reach admin pages

## How it works?

1. Admin logs in via /api/login
2. Flask checks 'role' field in DB
3. Admin redirected to /admin route
4. DB query returns ALL users & receipts
5. Full data displayed in dashboard

# Spending Analytic view




- 4 KPI cards: Total Spent, Tax, Count, Avg
- Line graph with smooth bezier curves
- Two datasets: Amount (pink) & Tax (light)
- Tooltips showing receipts per day
- Auto-refreshes after upload or delete
- Date-labelled X-axis, responsive layout

# Multi-Language Support

**EN English**
Default

**తె Telugu**
తెలుగు

**हि Hindi**
हिन्दी

**த Tamil**
தமிழ்

**മ Malayalam**
മലയാളം

**ಕ Kannada**
ಕನ್ನಡ

# Developments in milestone-4

## Chat Bot

An AI assistant inside dashboard to help users interact with their receipts

## Fully functional Admin Dashboard

## Dark theme for the website

## Bills digitiser

# Future Enhancements

## Chat Bot

A more advanced functioning chatbot

## Export Functionality

Export receipts to Excel, CSV, or PDF reports

## Landing page

A landing page to introduce the users about the application

## Advanced database

Azure blob or any secure platform for backend

# Thank You!

AI-Powered Bills Management System

Team- D

Presented By: Trishala, Sriya, Rithwik, Mastan and Atul